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**Brief Empirical Report**

# Utility of a Short-Form Phonemic Fluency Task

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## ABSTRACT

**Objective**: to establish a proof-of-concept and ascertain the reliability of an abbreviated 30-second (30s) phonemic fluency measure as a cognitive screening tool in older adults.
**Methods**: in all, 201 English-speaking individuals with normal cognition (NC; $n = 119$) or cognitive impairment (CI; mild CI or dementia; $n = 82$) were administered a standard 60s phonemic fluency task (FAS/CFL) with discrete 30s intervals denoted.
**Results**: for all letters, 30s trial scores significantly predicted 60s scores for the same letter, $R^2 = .7–.9$, $F(1, 200) = 850–915$, $p < .001$. As with 60s total scores, 30s cumulative scores (for all three trials) were significantly different between NC and CI groups ($p < .001$). Receiver operating characteristic analyses showed that 30s total scores distinguished NC and CI groups as effectively (AUC = .675) as 60s total scores (AUC = .658).
**Conclusions**: these findings support the utility and reliability of a short-form phonemic fluency paradigm, as 30s performance reliably predicted 60s/trial totals and was equally accurate in distinguishing impaired/non-impaired groups.

## INTRODUCTION

Verbal fluency (VF) tasks are among the most popular psychometric measures for assessing language functioning in adults (Faria et al., 2015). The two main types of VF tasks are phonemic (often called "letter") and semantic. Phonemic fluency tasks require examinees to recite as many words as they can beginning with a certain letter as quickly as possible. Semantic fluency (SF) tasks require individuals to name words belonging to a category, commonly animals or fruits/vegetables.

VF tests have proved useful as cognitive screening measures since they directly evaluate lexical access (e.g., speed of word retrieval), but additionally involve elements of executive function and other components of language. Levels and patterns of performance can aid in the differential diagnosis of various conditions such as Alzheimer's disease (AD) (Henry et al., 2004), mild cognitive impairment (MCI) (Bauer & Malek-Ahmadi, 2021), and Parkinson's disease (Torralva et al., 2015). Furthermore, discrepancies between performance on phonemic versus SF tasks may implicate the integrity of different brain regions as each involves overlapping yet different neural networks (Biesbroek et al., 2016; Meinzer et al., 2009). For example, research has shown a pattern of semantic greater-than-phonemic fluency decline in persons with AD (Vonk et al., 2020).

Most VF tasks are administered for 60s (with some exceptions for longer 90s, and several for shorter trials), with many examiners denoting discrete time intervals for word production throughout the task (e.g., as suggested in the administration manual for the Multilingual Aphasia Examination [MAE]). It has been demonstrated that response patterns on the task show an inverse relationship between time passed and answers given in clinical and healthy populations, with a majority of words being generated early (Crowe, 1998). However, little has been done to determine if the canonical minute length is particularly meaningful in a clinical sense, and whether shorter forms of the test may be just as useful. Moreover, in some situations, shorter tests that maintain the psychometric properties of the original may be even more useful in screening situations to reduce time and help to limit testing demands on patients.

In the context of VF tasks, short forms have been minimally explored. Hall et al. (2011) demonstrated that 30-second (30s) performance on an SF task (animal naming) was able to differentiate AD, MCI, and healthy groups. The authors showed this relationship held even for the first 15s of the task alone, wherein there were enough data to differentiate between various cognitive impairment (CI) groups. Similarly, Kim et al. (2011) demonstrated group differences in poststroke aphasic versus non-aphasic groups using an abbreviated SF task (animals). Regarding phonemic fluency, a single letter short form (letter "B") of the standard 60s length was found to be highly correlated with the more common three-trial "F–A–S" total score in healthy adults (Harrison et al., 2000). Furthermore, abbreviated fluency measures have been incorporated into popular cognitive screening instruments like the MoCA (letter "F") and the Frontal

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**Table 1.** Demographics and descriptive data by group

| | Total sample | Cognitive group | |
|---|---|---|---|
| | | NC | CI |
| Age ($M \pm SD$) | 68 ± 10 | 66 ± 9 | 72 ± 9 |
| Education ($M \pm SD$) | 14 ± 3 | 14 ± 3 | 14 ± 3 |
| Sex ($n$, %) | F (127, 63%)<br>M (74, 37%) | F (89, 75%)<br>M (30, 25%) | F (38, 46%)<br>M (44, 54%) |
| Race and ethnicity | | | |
| White | 53% | 50% | 57% |
| African American | 4% | 3% | 7% |
| Asian | 1% | 1% | 0% |
| Hispanic | 1% | 0% | 1% |
| American Indian | 41% | 46% | 35% |
| MMSE ($M \pm SD$) | 27.55 ± 3.1 | 28.8 ± 1.1 [26–30] | 25.7 ± 4.1 [10–30] |
| 30s scores F/C, A/F, S/L (total) | 9.0, 7.9, 8.4 (25.4) | 9.8, 8.6, 9.1 (27.5) | 7.7, 7.0, 7.4 (22.3) |
| 60s scores F/C, A/F, S/L (Total) | 13.5, 12.0, 12.9 (38.5) | 14.8, 12.8, 14.0 (41.6) | 11.7, 10.8, 11.4 (34.0) |

$M$ = mean; $SD$ = standard deviation; F = female; M = male; MMSE = Mini Mental State Examination.

Assessment Battery (letter "S"; Dubois et al., 2000), both of which use single-letter 60s trials for their fluency components.

The focus of this paper is on the validity and utility of short forms of phonemic fluency for older adults with and without neurodegenerative conditions. This investigation was conducted to fill the gap in the literature on short-form VF tasks: first, to determine if a 30s phonemic fluency score would be able to predict single-letter 60s score. Second, to evaluate if one abbreviated score could predict the total score across the three trials. Third, to compare short and standard forms to see how well they distinguish cognitively impaired and unimpaired groups to provide proof-of-concept and establish the preliminary reliability of a short-form, 30s phonemic fluency measure in older adults. Implications for its use in research and clinical practice are discussed.

## MATERIALS AND METHODS

Participants were obtained from a larger teleneuropsychology study of older individuals with and without CI (Cullum et al., 2014). Only fluency data that were collected during in-person test sessions were utilized for the present study. Participants included 201 English speakers ranging in age from 46 to 90 years, with education spanning 6–20 years. The University of Texas Southwestern Medical Center Institutional Review Board approved the study and all participants provided written consent.

The sample was divided into two groups: normal cognition (NC) and CI, with varying diagnoses determined via multidisciplinary consensus diagnoses through the Alzheimer's Disease Center. The impaired group consisted of 29 individuals who met clinical criteria for MCI (Petersen., 1999), 26 AD (National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association guidelines), or 27 all-cause dementia. These were supported by neurologic examination, neuropsychological evaluation, neuroimaging, and/or blood panels as available. Demographic information for the samples can be found in Table 1.

Phonemic fluency measures were administered by psychometrists or trained research assistants as part of a larger neuropsychological battery. Specifically, the Controlled Oral Word Association Test with phonemic stimuli of F, A, and S was used. This version is based on part of the MAE (Benton et al., 1976) which also contains an alternate form (C, F, L) for repeat assessment, which has been shown to correlate strongly (Ross et al., 2006). These letter sequences are also used in several other popular tests and sources (e.g., Delis–Kaplan Executive Function Scale; Delis et al., 2001, and Neurosensory Center Comprehensive Examination for Aphasia; Spreen, 1977). Their popularity has led to large sets of normative data to support scoring and interpretation (e.g., Heaton, 2004; Tombaugh et al., 1999).

The total number of words, excluding losses of set (e.g., proper names or stem words with different endings) and perseverations, uttered in 60 s made up the participants' scores. Examiners recorded responses for each of the two, 30s intervals; thus, the total number of words at 30 and 60 s were derived for each letter. For the current study, participants were counterbalanced to one of two conditions and their data comes from F–A–S or C–F–L.

Three primary analyses were conducted. First, a simple linear regression (LR) model was used to predict within-letter, or 60s letter total from the same 30s letter total, and this was done on all three letters for all examinees. Next, separate LR models were used to predict total VF scores (a summation of the standard 60s, three-letter totals) from a single, 30s letter trial. This was done to gage the feasibility of leveraging both methods of task-shortening, i.e., fewer stimuli or fewer trials, simultaneously. Finally, a receiver operating characteristic (ROC) analysis was done to compare the abbreviated and full-form's ability to distinguish CI and NC groups. All statistical analyses were conducted in SPSS version 28.0 and unless otherwise specified, were considered significant at the $p < .001$ level (two-sided).

**Table 2.** Single-trial 30s letter scores to predict (a) single-trial 60s letter and (b) total-task scores by group

| | | Model summary | | | ANOVA | | |
|---|---|---|---|---|---|---|---|
| | | *R* | *R* Square | std Error | *df* | *F* | Sig |
| (a) Single trial: abbreviated (30s) × standard (60s) | | | | | | | |
| F/C | NC ($n = 119$) | .900 | .810 | 2.118 | 1, 117 | 500.1 | <.001 |
| | Impaired cognition ($n = 82$) | .887 | .783 | 2.301 | 1, 81 | 297.4 | <.001 |
| | Total sample ($n = 201$) | .904 | .818 | 2.186 | 1, 200 | 897.3 | <.001 |
| A/F | NC | .906 | .820 | 1.934 | 1, 117 | 533.9 | <.001 |
| | Impaired cognition | .896 | .804 | 2.193 | 1, 81 | 331.5 | <.001 |
| | Total sample | .905 | .820 | 2.039 | 1, 200 | 910.4 | <.001 |
| S/L | NC | .843 | .711 | 2.581 | 1, 117 | 287.3 | <.001 |
| | Impaired cognition | .939 | .882 | 1.793 | 1, 81 | 607.4 | <.001 |
| | Total sample | .894 | .802 | 2.289 | 1, 200 | 800.4 | <.001 |
| (b) Single abbreviated (30s) trial × Total score—all three (60s) trials | | | | | | | |
| F/C | NC | .844 | .712 | 6.693 | 1, 117 | 288.6 | <.001 |
| | Impaired cognition | .815 | .665 | 8.130 | 1, 81 | 160.5 | <.001 |
| | Total sample | .844 | .712 | 7.303 | 1, 200 | 494.8 | <.001 |

Std Error = standard Error; *df* = degrees of freedom; Sig = significance.

## RESULTS

Participants (mean age = 68.5) had levels of education spanning 6–20 years (mean = 14.1), and were 62.6% female. The healthy ($n = 119$) and impaired ($n = 82$) cognition groups were both 50% White and had mean Mini Mental State Examination scores of ∼29 and 26, respectively. See Table 1 for expanded demographic information. Raw scores on the tasks across the sample are shown in Table 1. For all three letters, the overall regression using 30s scores significantly predicted 60s scores on the same letter, $R^2 = .7–.9$, $F(1, 200) = 850–915$, $p < .001$. See Table 2 for details. The predictors accounted for 75%–88% of the variance (SE between 1.85 and 2.4) in the models. The model equation, 60s scores = .585 + (1.437 × 30s score), was able to estimate total 60s scores (e.g., a person scoring 8 in their first block would be expected to score 4.1 in their second half) within 1.7 points on average for our sample.

30s C/F also significantly predicted 60s task total scores, $R^2 = .71$, $F(1, 200) = 494.8$, $p < .001$, in the entire sample (see Table 2). The first letter in each form was chosen (C/F), but the other four permutations yielded similar results. This prediction remained significant when looking at the NC and CI groups separately, as well. For CI, the model using 30s single letter scores to predict 60s task total scores, $R^2 = .67$, $F(1, 81) = 160.5$, $p < .001$, performed just as well. When comparing groups, 30s cumulative scores (i.e., from all three trials) between NC ($M = 27.5$, $SD = 7.8$) and CI ($M = 22.2$, $SD = 8.3$) were significantly different, $t(200) = 4.6$, $p < .001$. Importantly from a clinical perspective, the results of the ROC analyses showed that 30s total scores distinguished NC and CI groups as effectively (AUC = .675, CI: .599–.751) as 60s total scores (AUC = .658, CI: .580–.735).

Finally, an exploratory analysis was run to see if results aligned with previous literature in fluency tasks showing maximum word generation earlier in a trial with responses tapering off over time within a trial (i.e., comparing the first versus the second 30s epoch) and within the context of cognitive decline (Hall et al., 2011). Fully 98.8% (599/606) of all trials administered across groups in the 201-person sample contained more responses in the first 30s half of the test, consistent with previous reports.

## DISCUSSION

This investigation examined whether a shorter, 30s form of the phonemic fluency task would yield results similar to the traditional longer versions. Results showed that the 30s version of each letter (C/F) significantly predicted the full-length versions, accounting for 70%–85% of the variance in the outcomes. Furthermore, with only a single 30s trial, our models predicted overall phonemic fluency total scores (from three 60s trials) accurately and within an average of 6 points. Finally, as expected, the NC group performed significantly better than the CI on both standard (60s) and short forms of the task.

VF tasks are among the most commonly used neuropsychological tests because of their validity, ease of administration, and sensitivity to detecting CI in a variety of disorders. And while currently, the predominant form relies on three trials/letters with 60s administrations, these results show promise for an abbreviated task through its predictive validity of the widely used, full-length measure. The findings also support previous work by Harrison et al. (2000) who tested a single-trial (letter "B") administration in a healthy British population and showed strong test–retest reliability and a very high correlation ($r = .94$, $p < .0001$) with the standard long-form (i.e., three-trial) administration. Similarly, Hall et al. (2011) reported strong correlations between partial and full trial scores in their older sample using an SF task.

In the last two decades, major strides have been made in increasing the availability of brief neurocognitive screening tests with the goal of decreasing the test burden on patients and providing options when there are time constraints for assessment. Given the high demands for cognitive assessment in clinical

settings, reducing the length of tests and test batteries while not sacrificing the sensitivity of measures to detect and characterize CI may help extend provider resources. Furthermore, patient attention and motivation are finite capacities, and performance may be maximized with briefer measures, particularly with more impaired and/or less agreeable patients or when brief assessments may be adequate or preferred (e.g., in inpatient settings). In addition to the time savings, a short phonemic fluency task may also reduce frustration in patients who perform poorly with extended silences toward the end of the test.

In situations where patients are performing very well, when being used as a screening test, a cut-off may also be useful, stopping people that reach a given ceiling, or one standard deviation above the mean after 30s (in our sample that would be ∼13). In addition, a single 30s VF task could be easily incorporated into cognitive check-up visits in various settings (e.g., primary care) and examined over time to assess for change. The clinical utility of brief fluency tasks in these situations merits further investigation.

In terms of limitations, the focus of this work was older adults without CI and a combined group of individuals with MCI or dementia. The mean level of overall impairment was rather mild, and thus, whether findings will apply to those with greater impairment remains to be seen. Next, despite sufficient statistical power, future studies of brief fluency tasks would benefit from larger, more heterogeneous samples. The diversity of the sample benefitted from the number of American Indian participants, but the cohort was otherwise rather homogeneous, well-educated, and predominantly female. Replication and investigation in more sociodemographically diverse groups at various levels of CI are needed to interpret the results in broader contexts. Similar analyses of other VF tasks (i.e., other letters and SF) are also needed, and are in progress.

In conclusion, a 30s phonemic fluency score was highly predictive of the traditional full 60s score in a large sample of individuals with and without CI. Furthermore, a single 30s trial of one letter accurately predicted the three-letter (3-min) total score with good accuracy (i.e., within 2 points). Normative data are not currently available for 30s phonemic fluency tasks, but the current results provide initial data along these lines. Future work may expand upon this to increase the robustness of the formula to estimate a 60s score from 30s performances, potentially allowing for the use of existing normative data. Moving to shorter trials, or even a single, short trial, may be considered a viable option for clinicians, especially in situations where there is limited time or when examinee engagement may be at risk.

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## CONFLICT OF INTEREST

None declared.

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## AUTHOR CONTRIBUTIONS

Jack R. Kaufman (Data curation, Formal analysis, Investigation, Visualization, Writing—original draft, Writing—review & editing), Hudaisa Fatima (Writing—review & editing), Laura Lacritz (Writing—review & editing), and C. M. Cullum (Conceptualization, Funding acquisition, Investigation, Resources, Supervision, Writing—review & editing)

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